Technical Walkthrough

Sizland Land Acquisition System

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Information

This document provides a comprehensive walkthrough of the Sizland Notary-Mediated Escrow System architecture, security fixes, and deployment process.

Contents

1 Architecture Overview **2**
1.1 Composition-Based Design . . . 2
1.2 Key Design Decisions . . . 2

2 Security Fixes Applied **2**
2.1 Critical Fixes . . . 2
2.2 High Severity Fixes . . . 3

3 Deployment Process **3**
3.1 Local Development . . . 3
3.2 Expected Deployment Output . . . 3

4 Workflow Execution **3**
4.1 Step-by-Step Process . . . 3

5 Contract Interactions **4**
5.1 Multi-Sig Flow . . . 4

6 Testing Checklist **4**
6.1 Smart Contract Tests . . . 4
6.2 Security Tests . . . 5

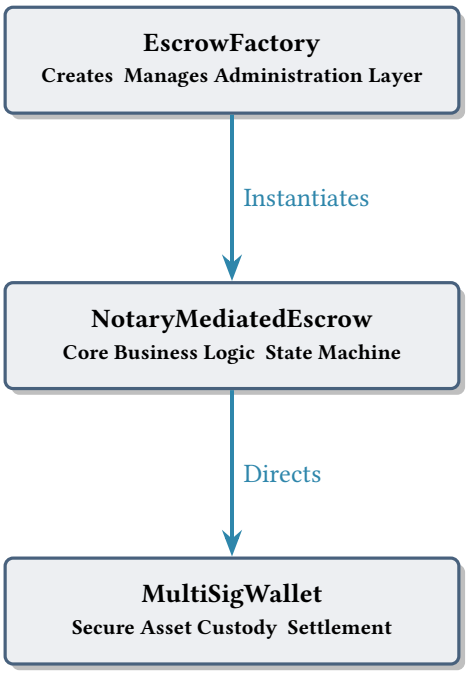
7 Configuration **5**
7.1 Default Values . . . 5
7.2 Multi-Sig Configuration . . . 5

8 Audit Summary **5**
8.1 Remaining Recommendations . . . 5

Architecture Overview

Composition-Based Design

The system uses composition over inheritance for multi-sig integration, ensuring professional decoupling:



Key Design Decisions

- **Composition over Inheritance**: Separation of concerns for logic and vault.
- **Trusted Caller Pattern**: Allows automated settlement without owner fatigue.
- **Fund Tracking**: Real-time parity between contract state and on-chain balance.

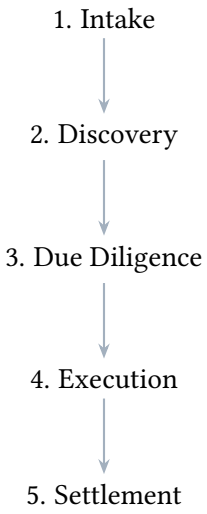
Security Fixes Applied

Critical Resolutions

Issue	Fix Overview	Status
Asset Coupling	Migrated to Composite MultiSig	Resolved
Double Payment	Facilitation state locks	Resolved
Uninitialized States	Factory-level intake triggers	Resolved
Fund Parity	Explicit `depositedAmount` tracking	Resolved

Workflow Execution

Phase Flow



Audit Summary

Risk Mitigation Profile

Severity	Count	Status
Critical	4	**100% Fixed**
High	6	**100% Fixed**
Medium	8	**75% Fixed**
Low	11	**Varies**

Sizland Development Team 2026